WORKED SOLUTION

STEP II 1994, Question 10

Pure • Collision

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q10

The key idea

Work with the *extension* of the rope rather than with the trailer's position. Because the truck's velocity is constant, the extension obeys exactly the same equation as the trailer's acceleration, up to sign, and that equation is simple harmonic motion. So the rope stays taut for half an oscillation, and then lets go of a trailer that is moving faster than the truck. After that it's a plain constant-velocity catch-up problem.

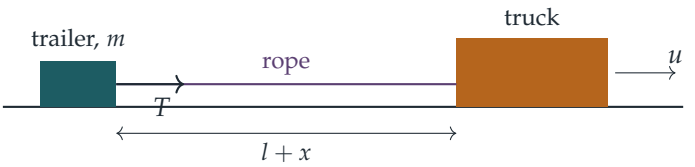
Solution

Setting up

Model the trailer as a particle, the rope as light and obeying Hooke's law while taut (and exerting no force while slack), and ignore friction. Measure time t from the moment the rope first becomes taut, and distances along the direction of motion. At $t = 0$ the truck is moving at speed u (and keeps that speed), the trailer is at rest, and the truck is a distance l ahead of it.

Let the trailer be at position $y(t)$, with $y(0) = 0$, so the truck is at $l + ut$. The extension of the rope is

$$x = (l + ut) - y - l = ut - y.$$



While the rope is taut

While $x > 0$ the tension is $T = \lambda x/l$, and it is the only horizontal force on the trailer, pulling it forwards. So

$$m\ddot{y} = \frac{\lambda x}{l}.$$

Since $x = ut - y$ and u is constant, $\ddot{x} = -\ddot{y}$, and hence

$$\ddot{x} = -\omega^2 x, \qquad \text{where } \omega^2 = \frac{\lambda}{ml}.$$

This is simple harmonic motion. The initial conditions are $x(0) = 0$ (the rope has just become taut) and $\dot{x}(0) = u - \dot{y}(0) = u$, so

$$x = \frac{u}{\omega}\sin\omega t, \qquad \dot{x} = u\cos\omega t.$$

The trailer's velocity is $\dot{y} = u - \dot{x} = u(1 - \cos\omega t)$.

> **Ask yourself** The SHM formula makes x negative after a while. Is it still valid then?

No. The equation $\ddot{x} = -\omega^2 x$ used the tension $\lambda x/l$, which only applies when $x > 0$; a rope can't push. The formula holds for $0 \le t \le \pi/\omega$, where $\sin\omega t \ge 0$, and the rope goes slack again at

$$t_1 = \frac{\pi}{\omega} = \pi\left(\frac{lm}{\lambda}\right)^{1/2}.$$

At that moment the extension is zero, so the trailer is again a distance l behind the truck, and its velocity is

$$\dot{y}(t_1) = u(1 - \cos\pi) = 2u.$$

After the rope goes slack

For $t > t_1$ the trailer moves at $2u$ and the truck at u, so the gap between them, which is l at $t = t_1$, shrinks at the steady rate $2u - u = u$. The gap is then less than l, so the rope stays slack, there is still no horizontal force on the trailer, and its speed really does stay at $2u$. The gap closes to zero after a further time l/u.

So the trailer collides with the truck at time

$$t_1 + \frac{l}{u} = \pi\left(\frac{lm}{\lambda}\right)^{1/2} + \frac{l}{u}$$

after the rope first becomes taut, as required.

Checks

- Units: λ is a force, so lm/λ has units of $\mathrm{m}\cdot\mathrm{kg}/\mathrm{N} = \mathrm{s}^2$, and both terms are times.

- Change of frame: the rope behaves like a spring in an elastic collision between the trailer and a truck of effectively infinite mass. In the truck's frame the trailer approaches at u and leaves at u in the opposite direction, so it ends with ground speed $2u$, just as found.
- A numerical integration of the trailer's equation of motion (with the force switched off whenever the rope is slack) for random values of m, l, λ, u reproduced the formula; for example m, l, λ, u chosen so that $\pi/\omega \approx 2.9944$ and $l/u \approx 2.3017$ gave a collision time of 5.2962 against the predicted 5.2961.

Common slips

- Applying the SHM solution for all time, forgetting that the rope goes slack when the extension returns to zero.
- Writing the equation of motion for the trailer's position y directly, which gives $m\ddot{y} = \lambda(ut - y)/l$, a forced equation. It's solvable, but the substitution $x = ut - y$ makes it unforced at once.
- Forgetting the final l/u: when the rope goes slack the trailer is still a full natural length behind the truck.

Beyond STEP

Changing to the truck's frame turns the whole episode into an elastic "collision" between the trailer and a spring attached to an immovable wall: the trailer comes in at speed u and leaves at speed u, and the contact lasts exactly half a period, whatever the speed. This is the simplest example of how Galilean invariance and energy conservation constrain collisions, an idea that runs through first-year mechanics and leads on to scattering theory. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both oscillations and collisions in this spirit.

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